



Using systems thinking causal loop modelling to analyze the water resources system in Rietspruit sub-basin, South Africa

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Abstract

A qualitative system dynamics modelling framework was applied in this study to analyze the main driving forces and their feedback interactions influencing water resources management in the Rietspruit sub-basin, South Africa. The dynamics of the water resource system were analyzed using the VensimTM PLE modelling platform and its archetypes of causal loop diagram, reinforcing and balancing loops. The model visually displayed the causal links of different processes influencing the water resource system and their feedback loops. Special focus was to ascertain and analyze the reinforcing feedback loops within a typical urbanized watershed, with the aim of depicting constraints that could influence sustainable water resource management and adaptation thereof. The causal loop model synthesized that, based on evidence from scientific literature and policy documents, the main driving forces towards the water resource challenges in the study area are associated with rising temperature, degraded water infrastructure and institutional and socioeconomic processes. The study, through the presented conceptual model, offered leverage points and potential pathways for sustaining both water quantity and quality, including infrastructure investment, sustainable revenue generation, decentralized groundwater use and public awareness and participation. The analysis in this study provided a conceptual framework for understanding of interactions of elements within the different systems influencing water resources in the Rietspruit sub-basin. This could be key for ensuring effective and sustainable management of the resource in the context of changing climate and socioeconomic developments.

Keywords Causal loop diagrams · Rietspruit sub-basin · System dynamics · Urban water systems · Water resources management

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Introduction

Human population growth coupled with climate shifts, urbanization and associated socio-economic changes, have resulted in water scarcity at a global scale and eventually posing serious threats to the development of countries (Javanbakht-Sheikhahmad et al. 2024; Guemouria et al. 2023; Wang et al. 2023). Water scarcity will remain a long-lasting challenge for humans in the years to come because as the demand for water for various purposes increases, the availability of fresh water is by and large declining, translating into a decrease in available freshwater per capita (Zanjanian et al. 2024). Presently, about half of the global population is experiencing severe water scarcity with varying time frames over the course of a year (IPCC 2023).

As urban and peri-urban areas continue experiencing an influx of population growth, including in informal settlements, it is also evident that there is an intensifying pressure on access to water and related sanitation services across different regions of the world (Sabale et al. 2022; Shemer et al. 2023; Chitonge 2020). Increasing population influences water resource availability and its quality directly through water use practices and indirectly through land use changes (He et al. 2021; Singh et al. 2025). Likewise, with continued global warming and increasing temperatures, the global water cycle is expected to be highly unstable, resulting in varying weather and climatic phenomena (Dinar 2024; Noh et al. 2024; Jain and Singh 2024b).

Changing climate affects the quantity and distribution pattern of water resources, hence it is challenging to effectively plan and allocate water to different users. Sector uses such as rural-urban migration, human health, ecosystem services, food security, industrial development and economic growth at large all hinge on water availability (UN-Water 2024). Thus, all these sectors are ultimately vulnerable to the impacts of climate change, indicating the need for adaptation of climate change through proper water management practices. A combination of climatic influences and socio-economic changes vis-à-vis population growth remains the constant and ever-present drivers to the challenges facing water resource management. To this effect, it is only natural that such ever-changing conditions be subjected to up-to-date institutional and economic approaches (Dinar 2024; Herman et al. 2020). Accordingly, it has been universally established that a holistic approach to water resource management is the feasible way to efficiently and sustainably manage water resources (Grison et al. 2023; Lenton 2011; Sugam et al. 2023; Nikolic and Simonovic 2015).

Managing water resources in different watersheds worldwide has been increasingly challenging due to a combination of various interrelated elements including socioeconomic factors, anthropogenic and natural changes

in the hydrological system and varying perspectives from diverse stakeholders and water users (Phan et al. 2021; Zanjanian et al. 2024). It has been argued by different researchers that to effectively and strategically plan and manage the sustainable use of water resources, it is vital that the knowledge of the dynamic interactions and feedback mechanisms of the related social, economic, climatic and other environmental factors be comprehensively understood (Elmahdi et al. 2007; Guemouria et al. 2023; Stojkovic and Simonovic 2019; Kotir et al. 2016). Nonetheless, the primary causes of the problems are not fully understood since the factors affecting water resource management are characterized by high degrees of complexities and uncertainties due to the different interactions and dynamics feedback amongst the multifaceted factors (Phan et al. 2021; Naeem et al. 2023). To overcome such complexities, the use of system dynamics modelling has been recommended because it is a holistic system approach that simultaneously models dynamic and non-linear interactions of the socioeconomic, as well as hydrological and environmental systems for effective water resources management (Naeem et al. 2023; Mashaly and Fernald 2020).

System Dynamics (SD) has been applied under different themes and objectives in water resource management, including analyzing impact of climate change on water resources and its adaptation measures, water allocation between different sectors, water supply and demand management, and stakeholder engagement for effective policy making (Guemouria et al. 2023; Perrone et al. 2020; Carnohan et al. 2020; Javanbakht-Sheikhahmad et al. 2024; Dzwauro et al. 2010; Chunga et al. 2023). In an attempt to solve complex water resource management problems, SD has been applied largely in two ways: firstly to understand the present status of water resource system at the watershed level to inform better planning process, and secondly to analyze how different changes in the strategies and/or conditions (e.g. policies, economy, climate) will impact water resources (Stojkovic and Simonovic 2019; Zheng et al. 2024; Kotir et al. 2016; Javanbakht-Sheikhahmad et al. 2024). Compared to conventional approaches such as the integrated water resources management, the SD approach is more desirable since it has the capability of showing how changes in the present state would affect the system in the future (Tian et al. 2020; Mirchi et al. 2012; Awah et al. 2024; Javanbakht-Sheikhahmad et al. 2024). Thus, in water resources, system dynamics is applied based on the notion that stakeholders in the water sector can better arrive at a decision with regards to developing and managing water resources depending on their level of knowledge of the water resource system components (Jain and Singh 2024a; Abdolabadia et al. 2018).

The use of qualitative system dynamics modelling (QSDM) to understand socio-hydrological systems is increasingly being highlighted. QSDM is used to map the complex web of interactions between the various elements influencing the dynamic water resource system and identify a wide range of policy interventions that are capable of reducing tradeoffs and increasing synergies between the various sectors (Giordano et al. 2025; Mirchi et al. 2012; Coletta et al. 2021). These interactions can be captured using causal loop diagrams (CLD), where the interlinkages between the various system elements are depicted. Moreover, based on anticipated feedback processes, CLDs provide the analytical ability to qualitatively evaluate system behaviour based on features such as reinforcing and balancing loops, time delays and non-linearity (De Paula et al. 2025; Groundstroem and Juhola 2021).

For example, Guemouria et al. (2023) employed the SD approach to understand the various interactions between water demand among different sectors with competing needs (domestic, agriculture, tourism, industrial) against the available supply in Souss-Massa basin (Morocco), where it was observed that the available policies could lead to unsustainable management of water resources. Employing the CLD component of the SD, Perrone et al. (2020) put forward different flood risk management policies in a coastal area in southern Italy through a collaborative problem identification and solution mapping process with key stakeholders. The CLD modelling was considered as a valuable tool for providing possible solutions to managing floods including increasing public awareness, proper land use management, and property damage insurance payout management. Bouchet et al. (2022) used the CLD to conceptualize the feedback mechanisms linking the social, ecological and hydrological elements of a groundwater system in the Nauru Island. By identifying the balancing and reinforcing feedback loops, the CLD showed that the social system had the greatest capacity to change the states of the system and it was the largest controller of the system performance. In India, Ram and Irfan (2021) also applied the systems thinking CLD approach to map and enhance the understanding of water related challenges amidst competing domestic, industrial and agricultural activities. Focusing on aspects of water supply and demand, water quality, water economy and its governance, the CLDs were used to capture the possible directions for sustaining water resources across the mentioned sectors (Ram and Irfan 2021). The conceptual system dynamic model has also been applied in Breede River Catchment (South Africa), where the challenges of water management and agricultural sustainability were holistically addressed (Nyam et al. 2021).

Recent advances have further demonstrated the value of integrated and climate-adaptive water resources

management frameworks for addressing the complex interactions among hydrological, climatic, socio-economic and agricultural systems. For example, Buri et al. (2024, 2025) developed a climate-adaptive optimal water resources management framework that integrates climate change projections with multi-sectoral water allocation to improve basin-scale resilience under future climatic uncertainties. These studies further reinforce the need for holistic decision-support frameworks capable of capturing complex system interactions, thereby providing additional motivation for applying systems thinking and causal loop modelling in the present study.

The Rietspruit sub-basin, the focus of the present study, has undergone far-reaching anthropogenic land use changes. The Rietspruit is a key tributary of the Vaal River, which is one of South Africa's most significant water systems supplying water for various uses in Gauteng and other provinces. The Rietspruit sub-basin and its surrounding catchment in the upper Vaal are facing serious environmental and socio-economic challenges affecting both water quantity and quality (Raji et al. 2022; du Plessis 2021; Obaid et al. 2023). Poor wastewater management due to insufficient maintenance of wastewater infrastructure has been persistent in the catchment (Sindane and Modley 2022b). Coupled with likely population growth, climate variability, and socio-economic developments, the water challenge issues could be compounded and aggravated if the present situation is left unattended. This could result in profound socio-economic challenges to the communities surrounding the Rietspruit and the wider Vaal River catchment. By and large, Vaal River's sustainability mainly relies on the sustainable management of its sub-basins including the Rietspruit (Dzwairo et al. 2024).

While system dynamics approaches, including CLDs, have been advancing the understanding and improvement of water resources management in other river basins in South Africa (e.g. Carnohan et al. 2020; Nyam et al. 2021), the Rietspruit sub-basin remains underexplored in this perspective. Therefore, the key contribution from this study is the development of a context specific CLD that captures the interplay of elements including pollution from mining and degraded wastewater infrastructure, urbanization and population growth and climate change. Such interactions reveal distinct water resource system behaviours and leverage points tailored to the challenges observed in study area. Analyzing and understanding the elements influencing water resource availability in the present times and beyond in the Rietspruit sub-basin is crucial for water resource planners and practitioners to devise sustainable approaches for managing water resources, including infrastructural planning and developments, financing mechanisms and water use strategies. Thus, this paper aims to assess such potential

drivers and pathways for future water resource changes in the Rietspruit sub-basin. Using the causal feedback loop diagrams of the system dynamics approach, the current research seeks to support the water resource stakeholders to incorporate current state and future information in their evaluation, planning, and development of policies related to water resource management.

Methods

Study area description

The Rietspruit sub-basin is a semi-arid area located in south-west of Johannesburg, Gauteng Province (South Africa), within latitude $26^{\circ} 21'$ and $26^{\circ} 44'$ South and longitude $27^{\circ} 33'$ to $27^{\circ} 56'$ East. The Rietspruit sub-basin covers an approximate area of 1139 km². The Rietspruit River, which is about 70 km long, flows from the north-eastern part of the sub-basin southward and is joined by three main tributaries (Fig. 1). Downstream the Rietspruit River near its outlet is the Vaal Barrage that is located on the Vaal River. Administratively, Rietspruit sub-basin is located within the Emfuleni Local Municipality (ELM) under the Sedibeng District Municipality (SDM) as presented in Fig. 1. ELM comprises approximately 80% of the total population in the SDM. The study area is exhibited by a gradual increase in residential land use particularly in the north-eastern part, covering both urbanized and informal settlements (Fig. 2).

As of 2022, the ELM had a population of nearly 950,000 with an annual growth rate of 2.62%. Almost 80% of the households in ELM had piped water connected inside their houses and 95% of total households had a flush toilet connected to sewerage (Municipalities of South Africa 2022). Correspondingly, the increasing population in the Rietspruit has resulted in increasing water users and demand annually (Sindane and Modley 2022b).

The average annual precipitation within the sub-basin is around 700 mm, whereby majority of the rainfall is received between November and April. Largely, the study area is dominated by agricultural activities, built-up area and gold and coal mining activities (Dzwauro et al. 2024; Banda et al. 2022). Agricultural activities in the upper Vaal are dominated by cattle grazing and dryland farming (Obaid et al. 2023), hence alterations in rainfall patterns and magnitudes are likely to increase the vulnerability of rain-fed farming. The land use/ land cover activities do not only influence water demand but also its quality, which has experienced extensive degradation (Sindane and Modley 2022b; du Plessis 2021).

Study approach

Model description

As is the case for many river basins, the broad elements influencing water resource management are the hydrological processes, environmental features, and socioeconomic

Fig. 1 Location of the Rietspruit River in relation to the Sedibeng District Municipality, South Africa

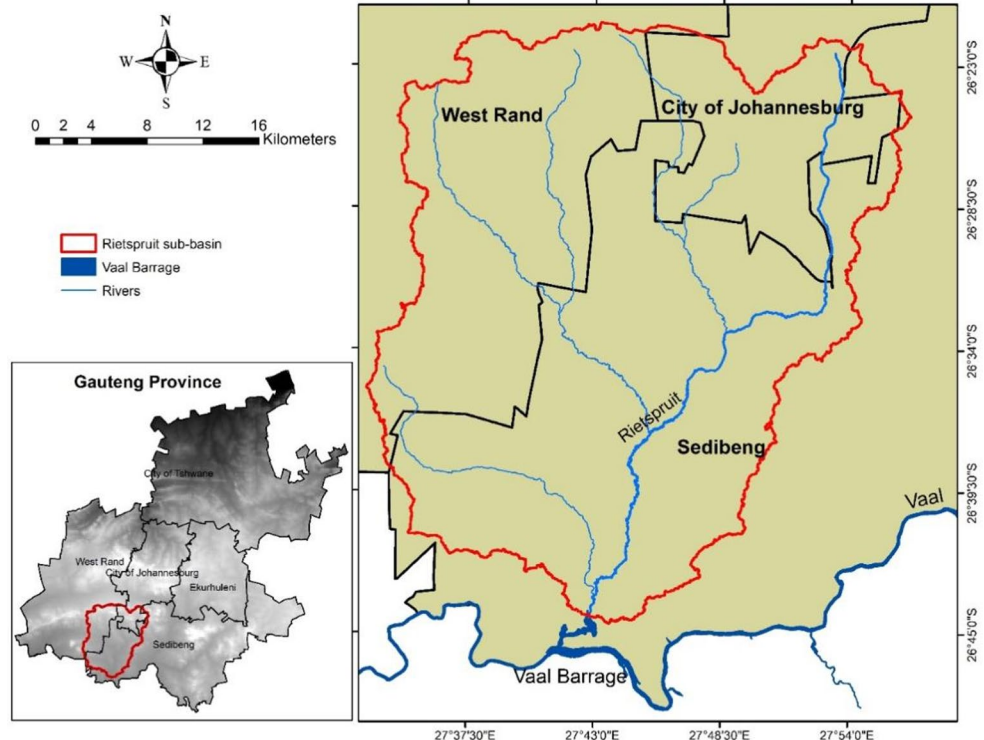
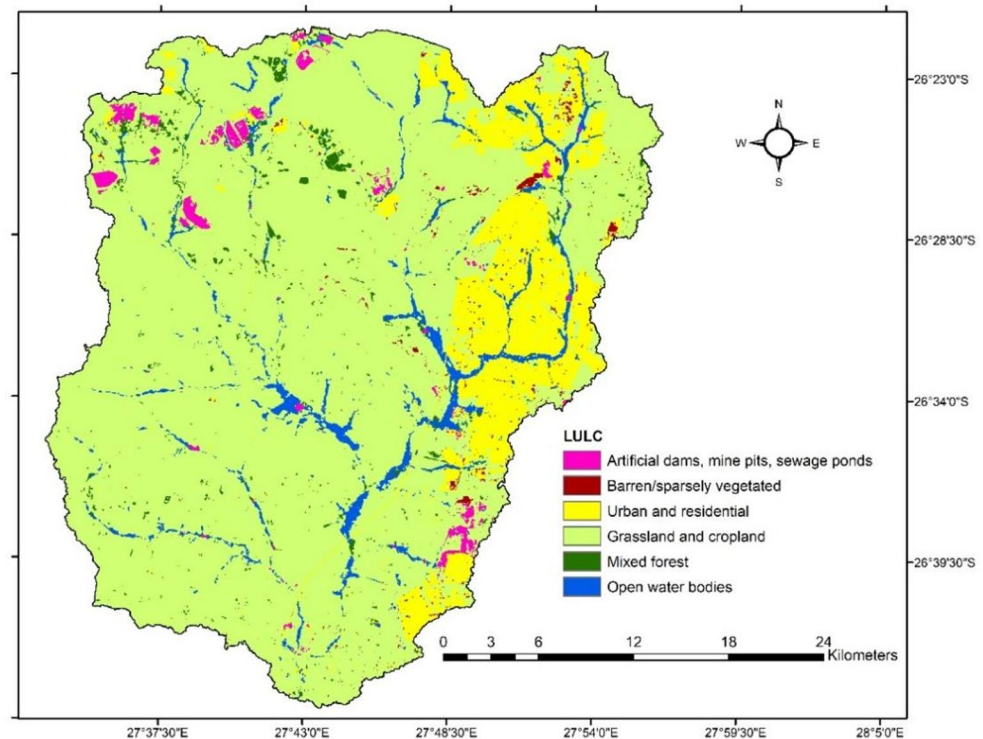


Fig. 2 Land use/land cover map for the Rietspruit sub-basin, South Africa



characteristics of the basin, which overall influence water supply, demand and long-term sustainability of the resource. Specifically, these include climate change and associated water supply, urbanization and population growth, infra-structural changes and economic development, changing consumption patterns and related water management policies (Bavandpour et al. 2021; Guemouria et al. 2023; Javanbakht-Sheikhahmad et al. 2024). Likewise, based on these generalized water resource management issues, a conceptual framework for the water resource system in the Rietspruit sub-basin is presented in Fig. 3. This model offers a framework for examining the dynamic behavior of Rietspruit sub-basin and pinpointing leverage points for successful policy intervention in light of population growth, economic development, and climate change. This conceptual framework depicts the complex interactions and feedback mechanisms found in a water resource management system, considering important drivers, supply and demand dynamics and policy interventions.

Causal loop diagram

This study adopted the CLD approach from the scientific point of view to pinpoint possible leverage areas and identify tradeoffs among varying strategies for addressing water challenges in the study area. In addition, from the mapping and visualization of the interlinkages between system components in the CLD, it is easier for the stakeholders including both experts and non-experts to understand the system,

eventually facilitating a broad discussion on the strategic planning and decision-making for water resource management in the present study area.

In reference to the presented conceptual framework (Fig. 3) and literature reviews, a CLD for the Rietspruit sub-basin was developed in this study. The present CLD expresses the interactions of different elements including, climate change, water supply and demand, economic development and population growth, which collectively influence water resource sustainability. The CLD has a feedback structure whereby the various model elements and their causal relationships are captured, thereby providing a comprehensive understanding of the water resource challenges exhibited in the study area. The CLD comprises arrows that connect different variables in the system, with each arrow indicating the influence of one variable on another variable.

Further, the CLD expresses the relationships between variables in positive (+) and/or negative (−) feedback. The positive feedback signifies that an increase (or decrease) in a certain Variable 1 will lead to an increase (or decrease) in a corresponding Variable 2. This indicates a positive correlation and such a loop is termed a reinforcing loop, presented as R (Guemouria et al. 2023). Contrary, a negative feedback shows an inverse correlation between two variables, whereby an increase in Variable 1 results in a decrease in the corresponding variable and vice versa (Javanbakht-Sheikhahmad et al. 2024; Jiang et al. 2020). Such a loop is described as a balancing loop, denoted as B. In essence, positive feedback loops produce growth and intensify deviation

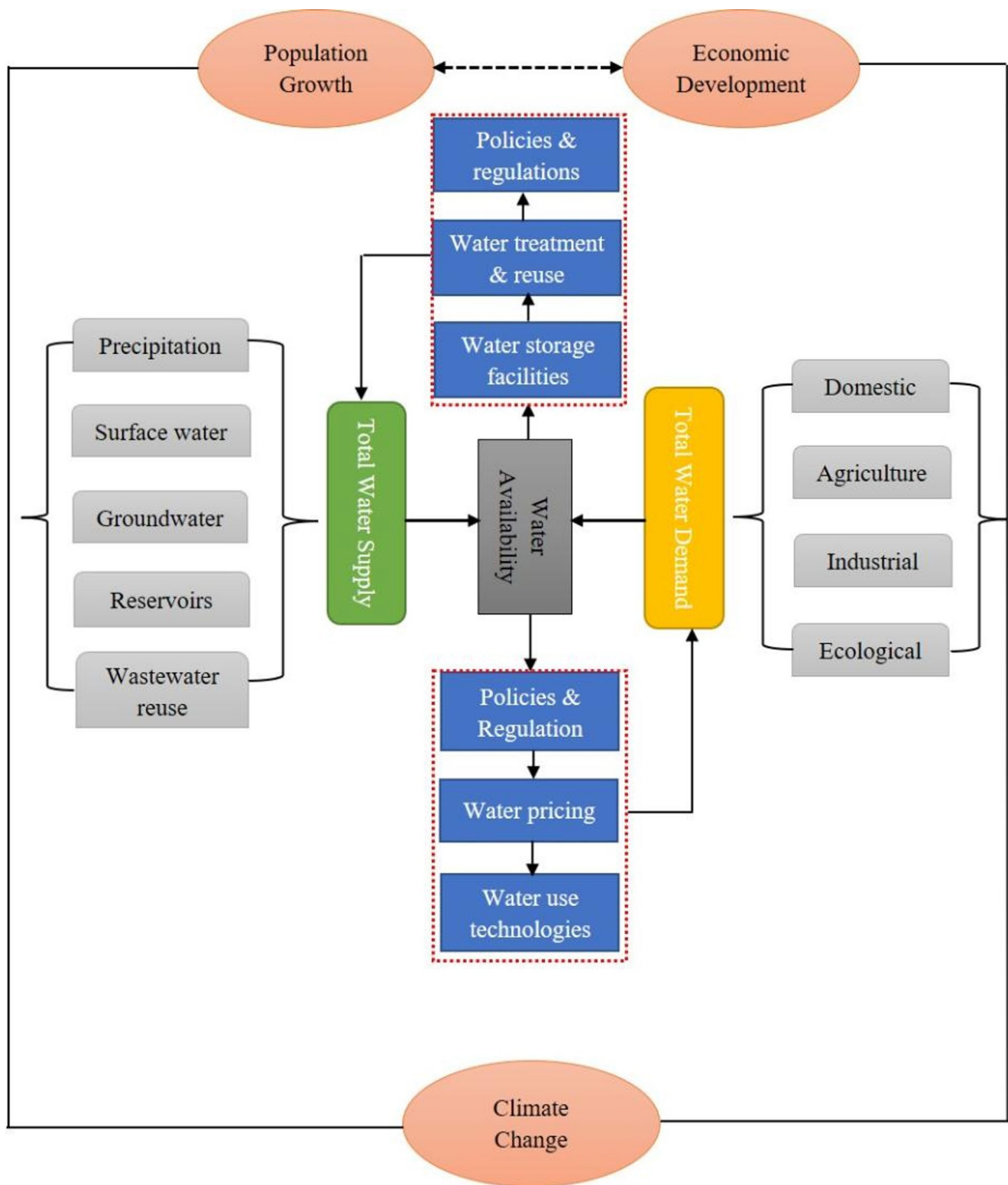


Fig. 3 Formulation of the model and analysis framework for the Rietspruit water resource system

in the system whereas negative feedback seeks to create a balance and equilibrium in the system, thereby bringing the system towards the desired state (Coyle 1996). The negative feedback works on the notion that whenever there is a discrepancy in the actual and desired states in the system, there is a need to initiate actions in order to lessen and/or eradicate the differences (Coyle 1996). Based on the complexity of the problem and its narratives, a CLD can have only a reinforcing loop, or balancing loop, or sometimes several reinforcing and balancing loops in it.

Overall, the CLD developed in this study is categorized into three main sub-elements: (i) water demand model, (ii) water supply model, and (iii) water governance model. Some of the common software tools for developing qualitative and quantitative SD simulation programs include Stella, Dynamo, Vensim, Powersim, and iThink. These models allow one to communicate how a system works—what goes into the system, how those inputs influence the system, and what are the outcomes (Balali and Viaggi 2015). The VensimTMPLE Version 10.2.2 Ventana Systems, Inc., software package was used to construct the CLDs, whose use is prevalent as it is an open source, user-friendly software for both visualizing and simulating real systems thereby enhancing their functionality (Treves et al. 2025).

Empirical grounding of model variables and causal links

The identification of system variables and their causal relationships was informed by an iterative synthesis of empirical evidence obtained from peer-reviewed studies, government reports, municipal planning documents, demographic statistics, hydrological and climatic analyses, water quality investigations, and field observations within the Rietspruit sub-basin and the broader Upper Vaal region. Accordingly, the CLD in this study integrates documented observations into a systems framework that illustrates how multiple biophysical, socioeconomic and governance processes interact through reinforcing and balancing feedback mechanisms. Table 1 summarizes key variables and their empirical bases that underpin the major components and causal relationships represented in the conceptual model.

Model validation

Unlike quantitative system dynamics models, qualitative causal loop modelling is not intended to reproduce historical time series or predict future system behaviour. Largely, CLD modelling aims to represent the feedback structure of complex systems and facilitating the understanding of systems. Thus, the model's credibility in the present study was assessed through structural validation and not behavioural validation. Structural validation involved ensuring that the

variables and causal relationships included in the CLD were consistently supported by peer-reviewed literature, government reports, municipal planning documents and field observations from the Rietspruit sub-basin.

The CLD developed in this study is positioned as an exploratory scoping model rather than a fully calibrated model. The CLD was developed through iterative synthesis of empirical studies, peer-reviewed literature, municipal documents and government reports in the Rietspruit and upper Vaal region, thereby ensuring logical consistency and agreement with documented processes affecting water resources in the study area. The study does not include the quantitative projection of the variables, parameter estimation or validation against time series data. The qualitative model will thus provide conceptual information and support dialogue by water resource stakeholders on sustainable water resource management in the study area, including how climate change, environmental alteration and socioeconomic developments—in the context of systems thinking—would influence future water availability and management.

Results and discussion

The findings presented in this paper focus on providing a high-level scoping conceptual model to explore various future scenarios regarding sustainable water resources management in the Rietspruit sub-basin. Being a qualitative model, the study emphasizes on the probable trends and feedback loops of the future scenarios rather than the actual values. The CLD that shows the key variables, and their interlinkages is visually presented in Fig. 4. Several reinforcing (R) and balancing (B) feedback loops that control the system's behaviour and sustainability potential are identified in the diagram. The positive (+) sign on the arrow indicates a positive relationship while a negative sign (−) shows a negative relationship. The double/cross lines cross in the arrow indicate the time delay in the response of a system component. The diagram depicts the different but integrated systems of water supply, demand and governance which interact with each creating a dynamic equilibrium in the water resource system, featuring a series of both reinforcing loops (R1 to R4) and balancing loops (B1 to B4).

Water supply and demand sub-component

In the Rietspruit sub-basin, the water supply subsystem largely concentrates on surface water resources due to small-scale use of groundwater. In the Gauteng Province, groundwater has received minimal attention such that its potential to augment bulk water supply is less understood (Water Research Commission 2019). Nevertheless, in this

Table 1 Summary of empirical evidence supporting the conceptual model

CLD variable/process	Empirical evidence from the Rietspruit sub-basin and its broader region	Key references
Population growth	Sedibeng population projected to increase from 1.06 to 1.13 million by 2029; top 10 highly urbanized municipalities in South Africa	Stats-SA (2024), Emfuleni Local Municipality (2022)
Urbanization	Expansion of built-up land in Rietspruit, with urban area increasing by 101% between 1986 and 2018; highly urbanized Emfuleni Municipality	Obaid et al. (2023), Emfuleni Local Municipality (2022), Banda et al. (2022)
Climate change	Approximately 2.2 °C warming (1992–2019); projected future warming of 3.35 °C	Banda et al. (2021), Banda et al. (2024)
Water quality deterioration	Pollution associated with untreated wastewater discharge; mining activities (uranium levels in water and sediment reaching 781.9 µg/L)	Tempelhoff (2021), Raji et al. (2022), Sindane and Modley (2022a), Dzwauro and Otieno (2014), Grange (2024)
Water scarcity	Projected Upper Vaal supply deficit of ~31% by 2030	Maluleke (2024), Prins et al. (2022)
Wastewater inflows to rivers	Stable dry-season flows maintained by treated effluent discharges	Banda et al. (2022)
Groundwater	Potential drought-buffering role recognised in national studies	Department of Water and Sanitation (2024), Water Research Commission (2019)
Agricultural adaptation	Increasing climate variability affecting dryland and peri-urban agriculture	Modibedi et al. (2020), Obaid et al. (2023)
Ageing infrastructure	Water losses, leakage and deteriorating infrastructure documented within Emfuleni Municipality	Tempelhoff (2021), Emfuleni Local Municipality (2022), Wegelin et al. (2009)
Governance challenges	Institutional, financial and operational constraints affecting water service delivery	Makaya et al. (2020), Grange (2024)
Infrastructure investment	Municipal plans prioritise rehabilitation and upgrading of water infrastructure	Tempelhoff (2021), Emfuleni Local Municipality (2022)
Demand management	Water conservation and demand management constrained by operational and maintenance deficiencies	Emfuleni Local Municipality (2022), Wegelin et al. (2009)

study, the role of groundwater in the overall management of water resources is recognized as an alternative water source to increase total water availability (Loop B2). This is also in consideration that groundwater and surface water must be managed as a single resource due to their interconnectedness.

Population growth is represented as one of the dominant drivers within the water demand subsystem. South Africa's population has been on the rise, exceeding 63 million people as of 2024 (Stats-SA 2024), whereby the Gauteng Province remains the most populous one, at an annual growth rate of 2.6% between 2011 and 2024. Further, it is anticipated that by 2030, the imbalance between water supply and demand will exceed 17% (Maluleke 2024). Specifically, the Sedibeng District Municipality (Fig. 1), which takes up the greatest area of the Rietspruit sub-basin, is projected to have its population increased from the current 1.06 million to 1.13 million by 2029 (Stats-SA 2024). Meanwhile, the population increase is largely attributed to the significant increase in life expectancy associated with socioeconomic

development and advances in public health, medicine and associated access to better health services (Stats-SA 2024).

Urbanization in the Rietspruit is also on the increase characterized by a clear surge in built-up land, with the highest increase in north-eastern part of the study area (Banda et al. 2022). Of the 205 local municipalities in South Africa, the Emfuleni Local Municipality (ELM)—which the study area is partly composed of—is one of the top 10 highest urbanized municipalities (Emfuleni Local Municipality 2022). Further, in the Gauteng region, the manufacturing industry is centered within the ELM. Thus, for years, economic growth and employment in the region have heavily been reliant on the manufacturing sector, which has topped the gross value added with agriculture contributing the least (Emfuleni Local Municipality 2022). These urbanization and economic activity scenarios point out that increasing population is likely to be a result of both rural-urban migration and natural growth.

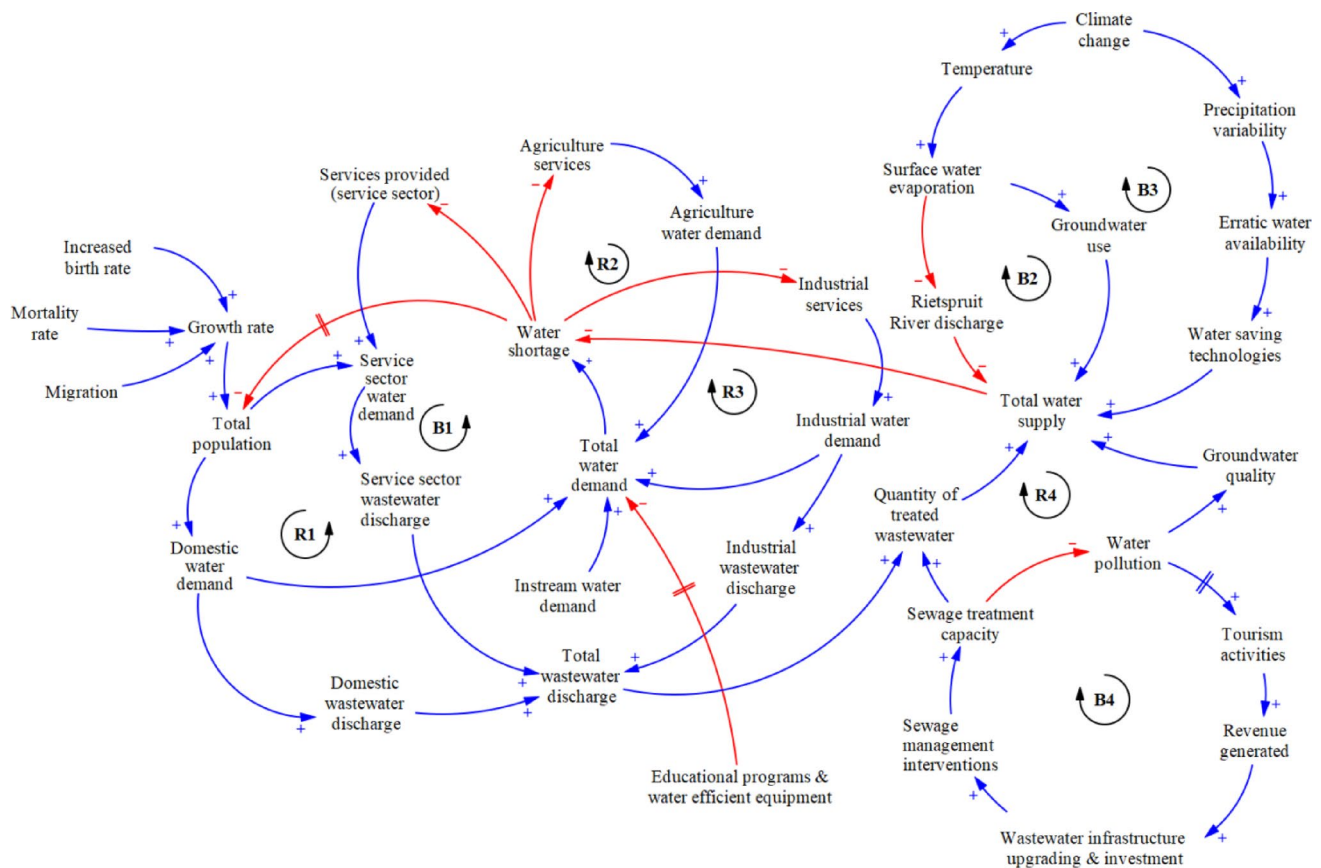


Fig. 4 Causal loop diagram shows visual depiction of the dynamic interactions and feedback mechanisms within the water resource system of the Rietspruit sub-basin

The observed and projected population increase, in addition to high economic developments and the resultant urbanization as indicated in Stats-SA (2024) and Emfuleni Local Municipality (2022), will likely put more pressure on the available water resources as the demand for domestic, industrial and agricultural water use. The CLD conceptually links these drivers via reinforcing loops R1, R2 and R3 (Fig. 4), hypothesizing that they could amplify pressure on water resources unless balanced by interventions. It has also been demonstrated that industrial activity, urbanization, and population growth raise the sub-basin's pollution load and water demand. The CLD suggests that water resource system could ultimately reach a tipping point where water availability and quality are seriously jeopardized if effective intervention is not implemented, as conceptualized by the reinforcing loop R1 and R4. An increase in water demand will reduce surface water availability and total water supply for varying uses, which will eventually reduce the total water demand. On the other hand, the CLD suggests increasing population and demand for water will increase the amount of wastewater that is available for recycling, thus potentially increasing water availability for reuse and consequently fulfilling water demand (Fig. 4).

Furthermore, studies have documented continued deterioration in surface water quality within the Rietspruit and surrounding Vaal system due to wastewater discharge, industrial effluents and mining-related pollution (Dzwairo et al. 2024; Sindane and Modley 2022b). Decreasing water quality will likely intensify the magnitude of water scarcity as the water becomes unfit for normal uses. Indeed, a region's water security is not exclusively a water availability issue, but quality too. The rising population and deteriorated water quality are directly related and contributing to the continuing water scarcity challenges, thereby impinging on the availability of essential services, including water and sanitation. The CLD represents these interacting processes through reinforcing loops R1–R4.

Over and above the increasing population and competing demand for water, empirical studies in the Rietspruit sub-basin and its surrounding regions have experienced erratic rainfall, rising temperatures and associated droughts. For instance, the Vaal Dam had water levels in 2024 lowered to below 35% as a result of prolonged dry spells (Oladele 2024). The low storage level illustrates the sensitivity of the regional surface water supply system to prolonged dry periods, providing empirical support for including climate

variability as a key driver within the conceptual model. Evidence from the historical climatic records in the Rietspruit shows that there has been a significant increasing trend in temperatures, with strong warming trends and temperature increases of nearly 2.2 °C in the period 1992–2019 (Banda et al. 2021). Furthermore, CMIP5 climate projections in the study area indicate that future maximum temperatures under RCP4.5 and RCP8.5 will also increase while precipitation is projected to either remain the same or experience a non-significant increase under different climate change scenarios (Banda et al. 2024). These observed warming trends support the inclusion of increasing evaporation as a mechanism reducing effective water availability within the CLD.

Geographically, South Africa is described as a water-scarce country, largely based on the unpredictable precipitation and high frequency of droughts (Donnenfeld et al. 2018). Overall, water demand in the Vaal region, including upper Vaal, has already passed its requirements for socioeconomic development (Ilunga 2017), such that water demand in the future will be met through inter-basin transfers (du Plessis 2017). In the upper Vaal, it has also been projected that the region will experience a water supply-demand deficit of about 31% by 2030 (Prins et al. 2022). The imbalance in the supply-demand has also been necessitated by the natural occurrence in the region where mean annual evaporation exceeds mean annual precipitation and streamflow is more sensitive to evapotranspiration (Banda et al. 2022). Clearly, the declining water supply will be aggravated in the long term as temperatures increase due to global warming. Thus, the occurrence and frequency of droughts/dry spells are major determinants influencing natural water security in the area.

Under such circumstances of sporadic water supply, water users resort to depending on groundwater particularly for domestic purposes, thereby contributing to meeting the water demand for the growing population. Despite South Africa's high reliance on surface water, empirical evidence shows groundwater remains an invaluable resource supporting economic development in many dry parts of the country as well as during extended dry periods (Department of Water and Sanitation 2024; Water Research Commission 2019). Considering that long-term climate projections from the study area suggest no significant decline in precipitation but increased temperatures (evaporation), the CLD suggests that groundwater could play a crucial role in sustaining water security due to its capacity to cushion drought conditions, as conceptualized in balancing loop B2—(Fig. 4). Unlike surface water, groundwater storage is less affected by extended drought and related evaporation due to changing climate. Groundwater systems are relatively well distributed, hence they have a potential to offer a sustainable and decentralized solution and adaptation to improving water

supply during climate change induced drought (IAH 2019). Even in regions where groundwater resources are limited, domestic use of groundwater can reduce the pressure on the municipal water supply that is dependent on surface water, particularly where the risk of water shortages is high due to climate change effects. Nevertheless, for groundwater supplies to fully contribute to climate change adaptation, the resource must be well protected from contamination arising from anthropogenic activities. The increasing water quality degradation from untreated wastewater discharge and mining activities in the Rietspruit has increased the vulnerability of groundwater to contamination which would eventually lower its demand. Therefore, the conceptual model suggests that protecting groundwater resources from contamination enhances the potential for alternative water resources and thus increasing total water availability (Loop R4). This underscores groundwater protection as both a water resource management and climate adaptation priority.

Further to the common surface water and groundwater sources, there is a need to strengthen unconventional water sources, including the use of treated wastewater. Wastewater addition in the Rietspruit water supply system has shown to increase the flow of the Rietspruit River. Historical Rietspruit river discharge records have shown no significant decline during the dry season in comparison to the rainy season (Banda et al. 2022). The observed stable streamflow during the dry season was attributed to increased wastewater discharge into the river, from domestic, industrial and other service sectors. In urban and semi-arid watersheds, it is common for streams to rely on wastewater discharge for base flow whereby such streams intentionally or unintentionally support ecosystem functioning all-year-round (Luthy et al. 2015). Under continued urban growth and increasing water demand, the Rietspruit River's dependence on effluent discharge is expected to become increasingly important, partially also due to the rising risk of dryness from escalating temperatures. In this regard, the CLD conceptually positions treated wastewater recycling in balancing loop 4 and reinforcing loop 4—(Fig. 4), suggesting it could enhance surface water availability while requiring careful management to avoid water quality trade-offs. These findings further highlight the need for investment in wastewater treatment infrastructure, stricter effluent quality monitoring and water reuse policies that maximize water availability while minimizing ecological and public health risks.

As earlier mentioned, precipitation in the study area has been projected to experience no significant changes in the future, while temporal variability of precipitation including intra-annual variability will intensify (Banda et al. 2021). Given the comparable amount of annual precipitation, agriculture is one sector to be affected as extreme weather events like dry spells and heavy storms will be common

due to such high intra-annual variability. The study area is dominated by dryland agriculture (Obaid et al. 2023), hence temporal climate variability will remain an issue of concern. Further, the area has several farmers involved in peri-urban community vegetable farming under irrigation (Modibedi et al. 2020), which is not only crucial for ensuring food availability but also generating income for the farmers. As the population increases coupled with urbanization, it is more likely there will be an influx of such urban community gardens resulting in increased demand for water and difficulties in satisfying such demands. Within the CLD, increasing climate variability reinforces agricultural water demand while simultaneously reducing the reliability of natural water supply. Accordingly, the use of decentralized water conservation measures—including constructing farm ponds to slow down runoff in dryland farms and rainwater harvesting systems such as rooftop capture for small-scale gardens—are recommended to increase adaptation for the urban farmers. These interventions could simultaneously strengthen urban water security, improve agriculture resilience and reduce pressure on potable water supplied by the municipality.

Overall, the demographic, climatic, hydrological and water quality evidence indicates that water resource challenges in the Rietspruit sub-basin arise from interacting rather than isolated processes. Population growth, urbanization, industrial development, climate warming and deteriorating water quality collectively increase pressure on conventional water resources, while groundwater development, treated wastewater reuse, rainwater harvesting and decentralized water conservation measures present complementary opportunities for enhancing system resilience. Rather than quantifying these interactions, the CLD synthesizes the available evidence into an integrated feedback structure that illustrates how multiple pressures may reinforce one another and identifies potential intervention points for improving long-term water security.

Water governance sub-component

Having realized the potential for present and future water scarcity and its interlinked challenges, there is a growing emphasis by researchers and policy makers on the need for humans to change the way they interact, use and manage water (Javanbakht-Sheikhahmad et al. 2024). This emanates from the fact that human activities and urbanization influence the water demand while at the same time potentially damaging the available supply. Indeed, improved water management must take into account the potential and reality of the changing socio-economic aspects including geopolitical changes and their implications towards achieving water security (UN-Water 2024).

In addition to the rising population and urbanization, water governance remains one of the critical issues with regard to the water management situation in South Africa (Makaya et al. 2020; Wegelin et al. 2009). To effectively adapt to climate change effects and ensure sustainable water security, it is necessary that there are enabling frameworks to guide decision making and policies for the efficient use and management of water resources. For instance, during climate related hazards like droughts, the governance element and the effectiveness of the governance systems are crucial for effectively determining risk reduction measures and sustainable water access (Makaya et al. 2020).

Largely, water related problems are attributed to a combination of increased competition for the limited resource and weak governance (Global Water Partnership 2000). The governance related challenges that hinder water management in South Africa touch on both social, political, technical and financial limitations, that overall lead to water insecurity. The Rietspruit sub-basin, as part of the Emfuleni Local Municipality, has not been spared from ageing water infrastructure, which has been an ongoing problem in the Municipality. The problem has caused huge water losses and contamination of water resources through wastewater infrastructure leakages (Emfuleni Local Municipality 2022; Wegelin et al. 2009). This puts the community at high risk of water and air borne diseases, besides the already caused socioeconomic distress among the residents and businesses in the area (Sindane and Modley 2022b). For long, water conservation and demand management issues in the area have been triggered by lack of maintenance in the water supply infrastructure and associated backlogs, poor operation of the system and poor water use practices of water users (Wegelin et al. 2009). Likewise, lately, the current crisis in the study area has been collectively attributed to poor governance in the Municipality (Grange 2024). Investment in new water infrastructure, upgrading and rehabilitation of aged ones has been in the plans of the Municipality (Emfuleni Local Municipality 2022). The CLD represents infrastructure investment as a balancing process (Loop B1) (Fig. 4), whereby improved infrastructure reduces water losses, increases effective water supply and moderates water scarcity pressures. The CLD illustrates that investing in such infrastructure and ensuring viable water supply will help meet rising demand and reduce scarcity but will require sustainable generation of revenues, through measures such as appropriate billing systems and water tariffs and reducing non-revenue water. However, financing instruments like water tariffs are complicated particularly in developing countries like South Africa, as there exists conflicting policy objectives, such as ensuring water access to all including the poor, and ensuring that there is financial sustainability of the services provided (Leflaive and Hjort 2020). While

sustainable revenue generation through appropriate tariffs, improved billing systems and reductions in non-revenue water is necessary to finance infrastructure rehabilitation, tariff reforms must simultaneously preserve equitable access to water, particularly for low-income households. As a result, strengthening governance requires balancing financial sustainability with social equity.

Nevertheless, due to backlogs in the infrastructure to be rehabilitated and financial constraints in the present study area, it is largely viable to focus on the demand side management, also considering that there will always be an increasing demand from population increase and urbanization. This is further necessitated in light of climate change, which will cause uncertainty and huge variability in water availability in the study area and surrounding region (Ilunga 2017; Jury 2016; Dosio et al. 2021; Banda et al. 2024). As earlier indicated, the study area is projected to experience increasing temperature trends thereby increasing the risk of severe dry spells and drought (Banda et al. 2021). Concurrently, the projected non-significant changes in precipitation will be affected by high inter-annual variability (Banda et al. 2024), requiring further investment in infrastructure to store water and promote water savings technologies (Loop B3 - Fig. 4). Policies that include drought contingency planning and investment in resilient infrastructure generate a balancing loop (B3) (Fig. 4) that reduces the negative consequences of climatic variability.

As is the case for many river basins, water resources in the Rietspruit sub-basin cut across local administrative boundaries. This increases the complexity amid competing demand for water for different uses, which would require collaborative demand management strategies amongst all the stakeholders. It was estimated that approximately 63% of water in the ELM is lost through infrastructure decay and improper use by consumers (Emfuleni Local Municipality 2022). In its plans, the Municipality seeks to enhance water conservation and demand management programs including leak detection, installing pressure reduction valves, and carrying out door to door awareness campaigns. The model has shown that public awareness and participation is another critical social component. As public knowledge of water issues rises, so does popular pressure on legislators to implement water-saving practices. This is shown in reinforcing loop (R2) (Fig. 4) in which civic involvement promotes sustainable consumption and better policy enforcement. Nonetheless, from the user perspective awareness alone may not be enough to encourage individuals to adopt water-saving practices and may take a long time to be realized. As such, municipalities could provide subsidies for businesses and residents in order to replace older, inefficient fixtures with water-saving models.

In the water resource system, water quality plays an equally important role as water quantity since it determines water usability and the health of ecosystems. In the upper Vaal including the Rietspruit sub-basin, one of the biggest concerns from the water resource perspective is the deteriorating water quality from the collapsing water service delivery infrastructure that has existed for more than two decades (du Plessis 2021). Low levels of wastewater treatment and continued discharge of untreated effluent from leaking sewer pipes are the main causes of poor water quality in the Rietspruit sub-basin. The degraded water quality results in economic water scarcity due to non-existent or inappropriate water infrastructure. It is often reasoned that a country's water resources security, in addition to its natural availability, hinges on its water infrastructure (bulk and retail distribution systems) and also on the existing regulatory and institutional framework (UN-Water 2024). Nevertheless, it is the regulatory and legislative environment that is the enabler and dictates the availability of infrastructure and performance of the institutions relating to water resource development and management (Van der Merwe-Botha 2009). What is clear thus far is that the water quality system is a multisector and dynamic issue. While mining, urban and domestic effluents dominate documented pollution loads in the Rietspruit (Sindane and Modley 2022a; Raji et al. 2022; du Plessis 2021), agricultural activities may contribute non-point source nutrients and to sediment load to the downstream areas. These could manifest in the overall degradation of water quality, thereby creating a vicious cycle unless countered.

The restoration of Rietspruit River water quality will not only bring economic benefits to the larger Vaal region but significant gains can be realized by the residents within the sub-basin and the local municipality. In the Rietspruit sub-basin, untreated or poorly treated effluent from dysfunctional wastewater treatment works has been a primary driver of deteriorating water quality. The Emfuleni Local Municipality is aware that the collapse of the sewer network has affected tourism, aqua marine, fishing and other sporting activities, which collectively bring economic rewards to the area (Emfuleni Local Municipality 2022). Central to balancing loop 4 (B4) is the potential for effective treatment of sewage and management interventions to halt water pollution cycle, thereby fostering positive feedback. Considering competing financial needs across different sectors, it is critical that the Municipality comes up with an assessment of the benefits the water infrastructure system restoration would bring, not only from the water-supply perspective but also the monetary and non-monetary benefits. The decay in water and sanitation infrastructure is slowing down investment and economic growth. Certainly, improved Rietspruit River water quality is hypothesized to attract economic

activities including tourism, which has the potential to generate revenue for further investments in the water infrastructure system (Loop B4 - Fig. 4).

Successful implementation of the infrastructural upgrades could reduce microbial loads, mitigate eutrophication and enable economic benefits (such as fishing and tourism), eventually feeding revenue back into maintenance and upgrades. Balancing loop B4 demonstrates how economic benefits from tourism can be reinvested in wastewater treatment, resulting in improved water quality and perhaps increasing total water supply, acting as a balancing factor against pollution. Hence, it is important to explain the biophysical environmental, climate change adaptation and socioeconomic benefits that may come from investment in water infrastructure and discharging treated effluent into the Rietspruit River. Such an understanding of the benefits is critical for lobbying public and political support, particularly in reference to funding.

Drawing on systems thinking principles, the most promising interventions to be targeted are balancing loops that focus on infrastructure upgrades, sewage treatment, effluent reuse and decentralized groundwater use. These will weaken reinforcing loops R1–R4 (e.g., demand amplification, rising temperatures and pollution cycles) by addressing root drivers like ageing infrastructure, untreated effluents.

Limitations of the study

Despite that the presented CLD provided an integrated conceptual understanding of the water resource system in the Rietspruit sub-basin, some limitations are acknowledged in this study. Considering that the study employed a qualitative systems thinking approach, the presented CLD represents the structure and feedback relationships within the water resource system rather than quantifying the degree of interactions of system elements or predicting future system behaviour. As such, the model should be interpreted as an exploratory framework for understanding system dynamics rather than as a predictive decision-support tool. In addition, the model was developed through an iterative synthesis of peer-reviewed literature, government reports, municipal planning documents and field observations from the Rietspruit and Upper Vaal region. Thus, the model was not calibrated or behaviourally validated using long-term hydrological, climatic, demographic or socioeconomic time-series data. The emphasis of the present study was on establishing structural validity by ensuring that the identified variables and causal relationships were supported by documented empirical evidence relevant to the study area. Notwithstanding these limitations, the study provides critical framework for understanding the complex interactions among climatic, hydrological, socioeconomic

and governance processes influencing water resources in the Rietspruit sub-basin. The model offers a foundation for identifying knowledge gaps, facilitating interdisciplinary dialogue and informing future quantitative system dynamics modelling, stakeholder engagement and policy scenario analysis aimed at strengthening integrated water resources management.

Conclusion

A qualitative conceptual model is developed in this study to improve understanding of the multifaceted issues influencing water resource sustainability in the Rietspruit sub-basin. The primary contribution of this work is the first application of systems thinking and CLD to the Rietspruit sub-basin, which enables the identification of leverage points that offers targeted pathways for sustainable water resource management in this water constrained basin. The study used systems thinking causal loop diagrams to ascertain the drivers that influence sustainable water resource availability in the Rietspruit, thereby developing an understanding of the holistic water resource system.

The conceptual model synthesizes empirical evidence regarding the sub-basin's specific key challenges including population growth, industrialization, urban migration, climate change (rising temperatures and variable precipitation), deteriorating water quality and complex governance issues—as deeply interconnected drivers affecting sustainable water management. Overall, this study demonstrates how systems thinking can be used to synthesize diverse empirical evidence into an integrated conceptual understanding of water resource challenges within the Rietspruit sub-basin. By linking demographic change, urbanization, industrial development, climate variability, deteriorating water quality, ageing infrastructure and governance processes within a common feedback structure, the CLD provides a systems perspective that complements conventional sector-based water resource assessments. The methodology and findings are applicable to similar urban watersheds, offering a transferable framework for water resource management. Consequently, this can facilitate sustainable water resource management in the study area by:

- Highlighting leverage points where interventions such as infrastructure investment/upgrades and sewage treatment upgrades, effluent reuse and source protection, decentralized groundwater use and public awareness can improve water resource sustainability, guiding policy and management strategies.

- Facilitating stakeholder engagement since the visuals in the CLD aids communication among stakeholders, fostering collaborative solutions.
- Supporting adaptation options by identifying convergent issues (such as climate change and water quality deterioration; industrialization and governance challenges) which would allow authorities to prioritize high impact leverage points thereby addressing multiple challenges simultaneously, and divergent issues (such as climate change and urbanization) for identifying tailor-made adaptation strategies.
- Laying a foundation for developing data-driven, policy-based scenarios through quantitative system dynamics modelling, thereby enabling practical management solutions.

While the CLD developed in this study provides valuable structural insight into the Rietspruit sub-basin's water resource system, future research should build upon the conceptual model by incorporating participatory stakeholder engagement to refine the feedback structure. Further work should involve translating this CLD into a quantitative stock-and-flow system dynamics model parameterized with observed hydrological, climatic, demographic, infrastructure and water quality datasets. Such a model would enable calibration, scenario analysis and evaluation of alternative management interventions, thereby providing stronger decision support for sustainable water resource management under future climatic and socioeconomic change.

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Data availability The data supporting the findings of this study are not publicly available, but can be made available upon reasonable request from the corresponding author.

Declarations

Conflict of interest All authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest or non-financial interest in the subject matter or materials discussed in this manuscript.

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